
Retention You Can Predict, Scope and Delete: a (μ, γ, T) Law and a Structural Deletion Guarantee for a Physics-Structured Memory Store

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Abstract

1 What a memory system forgets is set by policy: a decay factor, an expiry timestamp,
2 a learned forget gate. Such a policy states when an entry expires, not how fast
3 the stored value degrades, what moving the dial buys, or what remains once it is
4 removed. We answer all three on a physics-structured memory store — a damped
5 symplectic step of a learned Hamiltonian — whose retention policy is a budget
6 in three numbers: the stored direction’s spectral mass μ , the damping γ and the
7 temperature T . The dial has a computable optimum: retention half-life is non-
8 monotone in friction, one V-curve minimized at $\gamma_{\text{crit}} = 2\varepsilon\mu$ ($\propto \mu^{-2}$ overdamped,
9 a mass-independent floor underdamped), spanning ≈ 11 orders of magnitude in
10 μ^2 , holding on emergent (learned-MLP) as well as designed units, 3/3 seeds (dim
11 4, hidden 64, laptop-CPU), and reproduced by direct nonlinear rollout with the
12 argmin agreeing to 0.35%. Retention is scopeable: at $T > 0$ friction preserves and
13 temperature erases, so a localized friction hole is a memory vault, $107.77 \pm 4.78 \times$
14 designed and $106.1 \pm 5.0 \times$ law-referenced on emergent (3 seeds each), though the
15 field-versus-scalar contrast naming its mechanism is designed-only. Removal is
16 structural: under canonical placement — Blleloch & Golovin’s (FOCS’07) — store-
17 level deletion is exact, the post-deletion store byte-identical at every load measured
18 to the store that never held the item, under three stated conditions (budget $\geq$
19 n_{cells} , leak = 0, priority/attribute eviction; recency eviction excluded), on a
20 designed store of dim 3, capacity 8–64, no learning. We state what decay does
21 *not* buy: against an exact adversary a boolean TTL flag in the same store is as
22 detectable as physical decay, and the store stops answering before it stops leaking.

23 1 Introduction

24 Every deployed memory store already runs a retention policy: a decay factor, an invalidation
25 timestamp, a learned forget gate, a DELETE call. Forgetting, however, is the failure mode benchmarks
26 do not measure (Yang, 2026; Uddin et al., 2026), and a policy is not a law. One cannot read off
27 a trained LSTM the half-life of a stored value, the knob that sets it, or what remains once the
28 value is removed. We study a memory whose forgetting is a dynamical property of the store, not
29 a bookkeeping rule over it, so that three operational questions have computable answers: *what is*
30 *the retention policy, and where is its optimum* (§3.1); *can retention be scoped to one item* (§3.2); *is*
31 *deletion real, and what still leaks* (§3.3).

32 Physics enters only as the derivation apparatus. The store’s state (q, p) advances by a damped
33 symplectic step of a learnable Hamiltonian $H = T(p) + V_\theta(q)$: the damping $p \mapsto (1 - \gamma)p$
34 supplies forgetting, Langevin noise supplies temperature T , and items are optionally written as *atoms*:
35 localized wells superposed into one energy function a dynamical relaxation reads. Our reference

unit is the CLU (Causal Learning Unit), introduced as CHLU in Jawahar & Pierini (2026). Its core is exactly solvable: diagonalised in the mass-whitened Hessian, one damped step acts on each normal mode as a single 2×2 matrix fixed by $(\varepsilon\mu, \gamma)$, whose spectral radius is that mode’s retention (Anonymous, 2026, a theory note on which nothing here depends).

Contributions. (1) *A retention dial with a computable optimum* (§3.1–§3.2): retention is non-monotone in the dial and the optimum is predicted, not tuned — one V-curve minimized at $\gamma_{\text{crit}} = 2\varepsilon\mu$, so latch, overdamped register and underdamped working memory are three regimes of one curve, evaluated at two values of μ , not two laws, across ≈ 11 decades in μ^2 , on emergent as well as designed units, on two instruments. (2) *Scoped retention* (§3.2): at $T > 0$ friction preserves and temperature erases, so a local change to the dial confines one item’s coordinate, and a friction hole is a vault whose two constituent laws hold on both arms. (3) *A structural deletion guarantee* (§3.3), a composition and not a new deletion algorithm: the placement rule and its delete-time repair are Blelloch & Golovin’s (FOCS’07); new here is only the composition — a lattice packing certificate, decaying content whose decay provably commutes with deletion, one canonical energy function read by a dynamical relaxation — which makes the store’s state a function of its live set alone, plus the trilemma that prices lifetimes and an honest leakage statement. (4) *A three-state memory lifecycle* (PROTECTED $\rightleftharpoons$ ACTIVE $\rightarrow$ TRASH, 7/7 legs, kill-conditions committed before the verbs, one leg unexercised) — the consolidation and stale-entry face of the same store — shipped as a mechanics demonstration on a toy synthetic build: no value or benchmark number is claimed, and none was run.

What class of claim this is. Our evidence is a small designed store measured at laptop scale: not a deployed agent memory, not an LLM system, not a benchmark result. We report a mechanism with measured laws, not a system result.

Nomenclature, because “mass” names two opposed quantities. The inertial mass M_i is the learned diagonal of the kinetic term (larger $\Rightarrow$ slower); the spectral mass is $\mu_k^2 = \lambda_k(M^{-1/2}\nabla^2 V_\theta M^{-1/2})$ (larger $\Rightarrow$ shorter memory). Retention statements use μ ; a stored value lives on a near-flat coset direction; ε is the integrator step, $= 0.05$, never a tilt. Half-lives mean nothing without a read tolerance Δ and the thermal persistence length ℓ_θ , so we never quote an $n_{1/2}$ without its Δ and ℓ_θ/Δ , and never read a designed-versus-learned distinction off raw exponents. Results on designed testbeds are labelled verification of exactness; on learned or emergent potentials, evidence. Every $T > 0$ result requires the fluctuation–dissipation-consistent noise scale $\sigma_i^* = \sqrt{M_i T \gamma (2 - \gamma)}$ and a Newtonian kinetic mode. These laws govern the store; end-to-end performance additionally depends on the encoder, measured separately, φ -bytes ledgered.

2 Related work

Retention policies in deployed memory systems. Agent memories are external stores a model writes to, retrieves from and edits (MemGPT, Packer et al., 2023; Mem0, Chhikara et al., 2025), or bounded compressive state inside the network (Infini-attention, Munkhdalai et al., 2024). Their retention policies are scores and gates: a 0.995 recency decay factor in a ranking heuristic (Generative Agents, Park et al., 2023), an Ebbinghaus decay of memory strength (MemoryBank, Zhong et al., 2024), a learned per-memory expiration span (Expire-Span, Sukhbaatar et al., 2021), and a learned forget gate α_t in $\mathcal{M}_t = (1 - \alpha_t)\mathcal{M}_{t-1} + S_t$ (Titans, Behrouz et al., 2025) — the nearest published neighbour to the dial of §3.1. None states a half-life, a read tolerance or a temperature, and none prices what is lost. We benchmark against none of them; the contrast is structural, our (μ, γ, T) budget occupying the same slot with those three quantities attached.

Deletion, deployed and formal. Zep invalidates a contradicted edge with a timestamp and preserves the record (Rasmussen et al., 2025); Mem0’s DELETE removes a row. Byte-identity is the opposite design point, and it matters because soft-deleted vectors stay reconstructible from raw index files in graph ANN databases (Chakrabortii et al., 2026). A 2026 evaluation wave motivates the framing: production failures are predominantly forgetting failures while benchmarks measure recall (Yang, 2026), failures to forget outdated memory drive a large share of sampled recommendation errors (Uddin et al., 2026), deleted facts stay recoverable through retained correlated modalities (Wang & Zhang, 2026), and agentic unlearning must handle reactivation out of persistent memory (Wang et al., 2026) — the last two being why §3.3’s encoder-excluded scope is a scope, not a hedge: the store deleting an item is not the system forgetting it. Formally, *certified* removal is Guo et al.’s (2020) §2 Eq. (1), an ε condition with an unnumbered (ε, δ) relaxation immediately after; exact methods

delete by isolation, with cost and capacity formalisms in place (Bourtole et al., 2021; Ginart et al., 2019; Sekhari et al., 2021). We make none of those claims: ours is a store-level bit-exactness statement with the encoder excluded, coining no benchmark and no cost claim. Order-independent placement is not ours either — Snyder (1977) through Naor & Teague (2001) to Blelloch & Golovin (2007), whose table we use; and Blelloch, Golovin & Vassilevska (2008) had already taken unique representation into computational geometry, so canonical placement in a geometric setting is not virgin ground (Appendix D).

Forgetting laws, learned and physical. Learned recurrent comparators (LSTM, Hochreiter & Schmidhuber, 1997; LEM, Rusch et al., 2022) lack a stateable forgetting law. In physics, dissipation turns type-A Nambu–Goldstone modes from propagating to diffusive (Minami & Hidaka, 2018) — exactly our coset — and a recent preprint on symmetry-protected Lyapunov neutral modes (arXiv:2605.03338) proves that an exactly G -equivariant flow has *at least* $\dim(G/\mathcal{H})$ zero Lyapunov exponents along the orbit: the kinematics of protection, to which our budget is the constitutive complement, since the gap is μ^2 .

3 Results

3.1 The retention dial has a computable optimum

If damping is the retention dial, where should it be set? Not by tuning, and not at the low end: retention half-life is non-monotone in friction, and the turning point is fixed by the stored direction’s spectral mass, so the optimum is predicted.

The step is conformally symplectic, $\det J = (1 - \gamma)^d$, and $(1 - \gamma\phi(q'))^d$ when position-gated; checkpoints (dim 4, hidden 64, $\varepsilon = 0.05$, 150 epochs) are *designed* ($SO(2)$ -invariant V_θ , 5 seeds) or *emergent* (MLP V_θ on identical symmetric data, 3 seeds). On a trained designed- $SO(2)$ vacuum the massive radial mode’s $n_{1/2}(\gamma)$ is a V minimized at $\gamma_{\text{crit}} = 2\varepsilon\mu_{\text{rad}}$, the overdamped branch following the μ^{-2} law up to $\varepsilon\mu \approx \gamma/2$ and then saturating at a mass-independent floor (log-slopes -1.006 and $+1.23$ – $+1.27$; 5 seeds, verification, Appendix A). The flat coset (Hessian $\mu^2 \approx 10^{-15}$) is the *same curve* at $\mu \rightarrow 0$, where $\gamma_{\text{crit}} \rightarrow 0$: latch, overdamped register and underdamped working memory are three regimes of one curve, evaluated at two values of μ — not two laws. It generalizes to emergent units (evidence): on three emergent MLP checkpoints trained on the same symmetric data the coset-tracked $T = 0$ half-life traces the same V-curve — argmin $0.902 \pm 0.003 \times \gamma_{\text{crit}}$, log-slopes -1.0020 ± 0.0003 / $+1.116 \pm 0.011$, 3/3 seeds. Across both families the curve spans $\mu^2 \in [1.7 \times 10^{-12}, 7 \times 10^{-2}]$, that low endpoint being the ring-profile probe’s resolution on a checkpoint whose Hessian μ^2 is machine zero rather than a spectral mass — so eleven orders is one curve on one instrument, though the $\mu \rightarrow 0$ corner has its own model-side bound of $n_{1/2} \geq 9.5 \times 10^{15}$ steps by direct $T = 0$ rollout (Appendix B).

Two instruments, one law (evidence). Every point above is a one-step Jacobian, so we re-measured the curve by direct nonlinear rollout, reading $n_{1/2}$ from the decay envelope’s rate (3 emergent seeds, $\delta = 0.05$ – 0.5 rad). The shape reproduces: argmin 0.9001 ± 0.0052 against the Jacobian’s 0.9032 ± 0.0027 in units of γ_{crit} — a 0.35% difference — log-slopes -0.983 ± 0.013 / $+1.117 \pm 0.010$ on identical fit windows, and decay *rates* agreeing to 0.995 ± 0.003 in the asymptotic window. The instruments differ by a level, not a law: a first-crossing threshold reads short by a per-checkpoint constant ($1.33/1.80/1.82\times$) flat in γ , with coefficient of variation 2.2–3.7%. *What would have falsified the law:* a stored direction that mixes normal modes, or is anharmonic at the write amplitude, has no single-mode reduction, so the curves would neither collapse nor put their minimum at $2\varepsilon\mu$; neither happens at ten times the write amplitude, where the argmin moves by at most 1.4% (to 0.897 – $0.917 \gamma_{\text{crit}}$) and the slopes by at most 1.9% (Appendix B).

3.2 Scoping retention to one item: friction preserves, temperature erases

Scoping a policy to one entry means changing the dial locally, and the sign of the dial’s effect on a stored value is not the expected one. At $T = 0$ a designed coset latches forever at any friction (drift $\leq 4.9 \times 10^{-12}$ rad over 200k steps, all $\gamma \in [0.002, 0.5]$, 5 seeds, verification): friction has no contractive action on a flat direction. At $T > 0$ the coset decays diffusively and friction slows it down — $D_\theta = \varepsilon T(2 - \gamma)/(2F^2\gamma)$ is verified to 1.0068 ± 0.0219 over 25 (γ, T) cells, so $n_{1/2} \propto \gamma^{+1}$ and the sign flip $\partial n_{1/2}/\partial \gamma > 0$ holds in 10/10 seed $\times$ temperature conditions, $\gamma : 0.05 \rightarrow 0.2$ *lengthening*

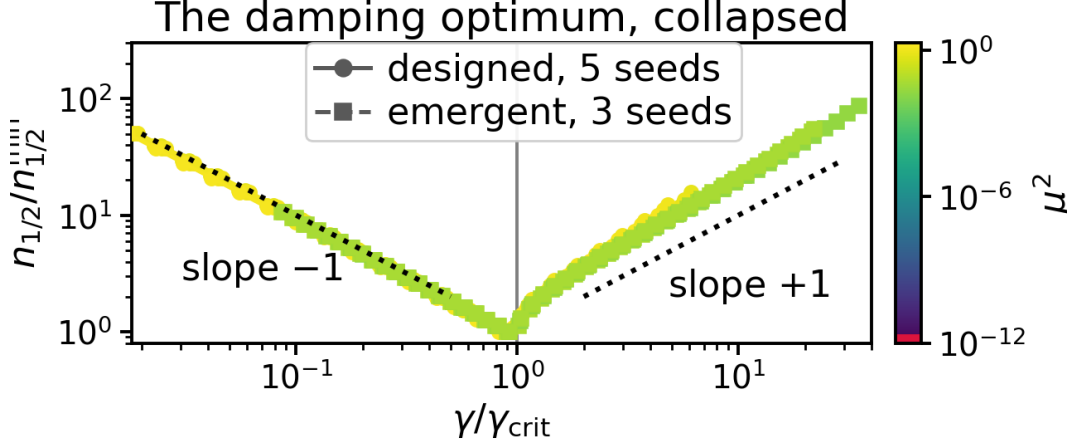


Figure 1: *The retention dial and its optimum, collapsed (headline)*. Eight measured curves collapse onto one: five designed radial modes (circles, verification, $\mu_{\text{rad}}^2 = 0.670\text{--}1.348$) and three emergent learned-MLP coset modes (squares, evidence, $\mu_{\text{soft}}^2 = 2.0\text{--}5.4 \times 10^{-2}$, 3/3 seeds), against the ∓ 1 asymptotes; the designed flat coset is the $\mu \rightarrow 0$ corner of the same curve. (*One-step Jacobian; dim 4, hidden 64, $\varepsilon = 0.05$, laptop-CPU; langevin_noise="fdt", Newtonian kinetic mode.*)

142 the half-life $3.77 \pm 0.23 \times$ at $\Delta = 0.5$ rad, $\ell_\theta/\Delta < 0.05$; only temperature has $\partial n_{1/2}/\partial T < 0$. *Fine*
 143 *print, beside the claim:* every $T > 0$ number requires langevin_noise="fdt" and a Newtonian
 144 kinetic mode; under the reference default these laws fail, and in the relativistic mode no noise scale
 145 targets Gibbs.

146 *Hence a memory vault.* The shipped position-gated friction field is *absorb-only*, so a hole is a brake
 147 and a refrigerator ($T_{\text{local}} = 1.26 \times 10^{-4}$ against 10^{-3} outside, where a coupled bath predicts none),
 148 and the vault factor is $(\gamma_{\text{eff}}/\gamma)^2 = 107.77 \pm 4.78 \times$ (D_θ -estimator, pre-registered, 3 designed seeds,
 149 $\Delta = 0.5$ rad) against $13.28 \pm 0.12 \times$ for a scalar friction of equal γ_{eff} , the ratio 8.11 ± 0.37 being
 150 the refrigerator. At $T = 0$ it erases nothing. Both laws transfer to an emergent register (evidence,
 151 3 emergent seeds at $T = 4 \times 10^{-3}$ and 8×10^{-3} , both above T^*): the refrigerator law holds at
 152 0.9998 ± 0.0019 of prediction (the coupled bath rejected again at 0.2235) and the $\hat{D}_\theta \propto \gamma_{\text{eff}}^{-2}$ law
 153 at $1.016\text{--}1.103$, a law-referenced vault of $106.1 \pm 5.0 \times$ beside the designed $107.77 \pm 4.78 \times$. One
 154 result has no designed analogue, and it is the one that makes scoping operational — *the hole confines*:
 155 the fraction of states outside the register ($|\theta| > 1$ rad) falls from $5.5/43.0/2.4\%$ with no hole to
 156 0.0000 inside it, 3/3 seeds, while a scalar friction of the same γ_{eff} still hops ($0.73/10.2/0.26\%$). *The*
 157 *contrast number stays designed-only*: on emergent the field/scalar ratio measures 23.39 ± 10.06
 158 because the *control* arm delocalises there ($\hat{D}/D_{\text{flat}} = 1.18\text{--}5.23$) while the field arm sits on the
 159 law, so 8.11 ± 0.37 is a designed quantity and the direction, not the number, transfers. We quote no
 160 emergent σ_θ ratio: above T^* that diagnostic is non-stationary here, its control returning 0.459 ± 0.118
 161 where it must return 1.000 (Appendix C).

162 *The designed-symmetry precondition, the boundary on everything above* (evidence; 3 emergent
 163 seeds and a matched designed control). The exact $T = 0$ latch is a *designed* object: on an MLP
 164 potential trained on identical symmetric data the softest direction is a mid-spectrum massive mode
 165 ($1 - |\lambda_{\text{coset}}| \approx 10^{-3}$ against the designed $\leq 1.1 \times 10^{-15}$), a written value relaxes completely,
 166 and capacity is $\approx 1\text{--}1.6$ bits. An emergent unit of this class has no continuous coset register, and
 167 continuum flat directions must be designed in. What survives is exactly the headline — the V-curve
 168 on 3/3 seeds and the sign flip above a measured $T^* \approx 3 \times 10^{-3}$. The same boundary bounds an
 169 extrapolation we do not make, and Appendix E records it.

170 3.3 Deletion: a structural guarantee, priced lifetimes, and what still leaks

171 (*Designed atom store, dim 3, no learning anywhere $\Rightarrow$ verification.*) Deletion in a deployed store is
 172 best-effort: a row dropped, an edge timestamped, a tombstone written. We ask instead whether the
 173 store's state can be made a function of its live set alone. Stated once: external benchmarks won on

174 their own headline metric = ZERO. A flat table deletes exactly by construction — the trivial substitute;
 175 at issue is a store whose items are *superposed into one energy function*, where placement would
 176 ordinarily depend on which others are present. Placement here is canonical, a function of the live
 177 records and the geometry alone, so removal reproduces the store holding the rest bit for bit. Measured:
 178 “with the waitlist, store-level deletion is exact at every load measured, $0.29 \times -1.71 \times$ lattice capacity
 179 — AUC 0.5000 ± 0.0000 on all six statistics, byte-equal 1.0000 — under three stated conditions
 180 (*budget* $\geq n_cells$, *leak* = 0, priority/attribute-based eviction; recency-based eviction remains
 181 intrinsically history-dependent and is excluded).” The allocator it replaces leaks membership after
 182 eviction (AUC 0.99985 at full load), and the packing price of exactness is negative: canonical
 183 placement admits strictly more keys than refuse-and-relocate. This is a store-level guarantee only
 184 — the frozen encoder and any residue of past writes in a learned landscape are separate channels,
 185 measured separately, and we claim no certified (ε, δ) unlearning; without the waitlist, exactness fails
 186 at overflow (Appendix D).

187 *Attribution, in the same breath as the theorems. Our placement rule is the Blleloch–Golovin*
 188 *(FOCS’07) stable-matching table, our Theorems 1–2 follow from theirs, and the fix-up cascade*
 189 *is theirs.* The composition’s price is negative where the general case can cost an exponential slow-
 190 down (Buchbinder & Petrank, 2003). That separation is between the *weak and strong* notions for
 191 comparison-based structures, whereas ours is against a history-dependent stochastic relocation rule,
 192 so the two statements are adjacent rather than identical.

193 *Lifetimes are priced by a trilemma* — exact value fidelity, amplitude-independent address hold,
 194 amplitude-independent read latency: pick two. The corner to lead with: *dropping amplitude-*
 195 *independent latency is the compute-adaptive-read dial — a faded memory costs more integration*
 196 *steps to read, a physical, measurable statement a timestamped row cannot make.*

197 *What decay does not buy.* The tighter laundering control is the policy already deployed elsewhere,
 198 a boolean TTL flag *inside the same store*, and it fires: against an exact adversary the TTL flag
 199 stays within 0.017 AUC of physical decay (0.983 versus 1.000), the two separating only against
 200 a resolution-limited adversary (0.559 versus 0.996 at $\sigma_{\text{obs}} = 0.1$), whose resolution is our own
 201 modelling choice. *The store stops answering before it stops leaking:* at the last amplitude before
 202 self-eviction ($A = 0.051$) retention is 0.832 — placement-dependent, quoted for the controller-placed
 203 disk — while a query-level membership adversary sits at AUC 0.983 and a white-box one at 1.000;
 204 the curves never cross. Against an *exact* adversary decay reduces the effect size, not the AUC, so
 205 we claim no reduction in distinguishability per se. What physical decay buys over a TTL flag is
 206 retrieval geometry, not privacy, and that differentiator needs no adversary model: *decay contracts the*
 207 *addressing tolerance of an item, $R_{50} 1.146 \rightarrow 0.752$ ($1.52 \times$), where a TTL dict’s lookup radius is a*
 208 *constant hard step at 0.75–0.77, independent of age.*

209 4 Limitations and future work

210 *Limitations.* (i) One architecture, one scale: dim 4, hidden 64, laptop-CPU, 5 designed / 3 emergent
 211 seeds; store at dim 3, no learning; lifecycle on a toy synthetic build. (ii) The designed-symmetry
 212 precondition (§3.2). (iii) Raw T/γ -exponents do not discriminate designed from emergent; the $\text{fdt} +$
 213 Newtonian scope is mandatory at every $T > 0$. (iv) The V-curve’s shape is instrument-independent,
 214 its absolute step counts are not (a first-crossing threshold reads $1.33\text{--}1.82 \times$ short of the Jacobian,
 215 a constant per-checkpoint offset), and the emergent collapse spans only $2.7 \times$ in μ^2 across three
 216 seeds, so the eleven decades are one decade of emergent variation attached to the designed $\mu \rightarrow 0$
 217 corner. (v) Deletion is store-level only, under set-function eviction with recency eviction excluded,
 218 and no deletion-cost claim is made ($O(n)$ rebuild per operation; unrun). (vi) No task-level payoff is
 219 claimed. *Scale is a scope choice:* this short is sealed to laptop-scale evidence; no larger-scale result
 220 is quoted or should be inferred. *Named next:* a wider emergent collapse in μ^2 ; the vault’s contrast
 221 arm re-measured where both arms stay bounded; amortized per-delete cost against the flat-datastore
 222 substitute; deletion at 10^3 -item stores; and a localized *temperature* field.

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A The (μ, γ, T) budget: the diffusion law and the sign flip

Designed $SO(2)$ testbed $\Rightarrow$ verification. All $T > 0$ cells require `langevin_noise="fdt"` and a Newtonian kinetic mode; the reference default is `legacy`, under which none of these laws hold. Every $n_{1/2}$ is quoted with its read tolerance $\Delta = 0.5$ rad and its ℓ_θ/Δ .

*Diffusion law, 25 cells ($5\gamma \times 5T$, seed 44): $\hat{D}_\theta/D_\theta^{\text{pred}} = 1.0068 \pm 0.0219$ (min 0.9644, max 1.0484); $d \log D/d \log T = +0.998, +0.997, +0.992, +1.010, +1.006$; $d \log D/d \log \gamma = -1.014, -0.980, -0.999, -1.001, -0.982$ after dividing the discrete $(2 - \gamma)$ factor. *Sign flip:* $d \log n_{1/2}/d \log \gamma = +0.9552 \pm 0.0422$ at $T = 10^{-3}$, 5 seeds, positive in 10/10 conditions; effect size $\gamma : 0.05 \rightarrow 0.2$ is $3.77 \pm 0.23 \times$ (per-seed 3.55, 3.64, 3.61, 3.89, 4.15); $d \log n_{1/2}/d \log T = -0.956$ to -0.979 deep-diffusive. *Latch at $T = 0$, designed only:* $||\lambda_{\text{flat}}| - 1| \leq 1.7 \times 10^{-14}$ over 22 γ values on 5 seeds, drift $\leq 4.9 \times 10^{-12}$ rad per 200k steps in 30/30 cells, against a $\gamma = 0$ control drifting 142.7 rad per 20k. *Designed V-curve:* argmin 0.076–0.096 against a predicted 0.082–0.116 (per-seed $\mu_{\text{rad}}^2 = 0.670, 0.771, 1.190, 1.093, 1.348$); asymptotic log-slopes -1.006 and $+1.23$ – $+1.27$; mass-independent floor 27.03 steps at $\gamma = 0.05$.*

Table 1: The $T > 0$ face of the budget, all four entries measured on the designed testbed.

stored direction	$\partial n_{1/2}/\partial \gamma$ (more friction)	$\partial n_{1/2}/\partial T$ (more heat)
coset / flat (designed)	> 0 , friction <i>preserves</i> ($+0.955 \pm 0.042$; $3.77 \times$ for $\gamma : 0.05 \rightarrow 0.2$)	< 0 , temperature <i>erases</i> (-0.956 to -0.979)
massive (underdamped)	< 0 , friction shortens (slope -1.006)	≈ 0 ; T sets only the fluctuation floor (≈ 27)

B The emergent arm, and the V-curve on a second instrument

Emergent MLP checkpoints, 3 seeds, plus a matched designed control $\Rightarrow$ evidence. Pre-registered; predictions filed before any harness ran.

*Four instruments, identical fit windows ($\gamma < \gamma_{\text{crit}}/2.5$, $\gamma > 2.5\gamma_{\text{crit}}$, keyed to the Jacobian’s γ_{crit} so the windows are not a free parameter): *I-J*, the one-step Jacobian, $n_{1/2} = \ln 2/(-\ln |\lambda_{\text{ret}}|)$; *I-R1*, rollout first crossing of $|\theta| \leq \delta/2$; *I-R2*, rollout last crossing; *I-R3* (primary), rollout envelope-rate fit over the late window, $R^2 \geq 0.9955$ (median 0.9996) on every cell.*

Finite write amplitude. At $\delta = 0.05/0.2/0.5$ rad the I-R3 argmin is 0.8928/0.9007/0.8968 (seed 42), 0.9047/0.9045/0.9174 (43), 0.9028/0.9037/0.9070 (44) — at most +1.4% over a $10 \times$ range — and the slopes move by at most 1.9%; every $\delta = 0.5$ write relaxed all the way home ($|\theta_{\text{final}}| \leq$

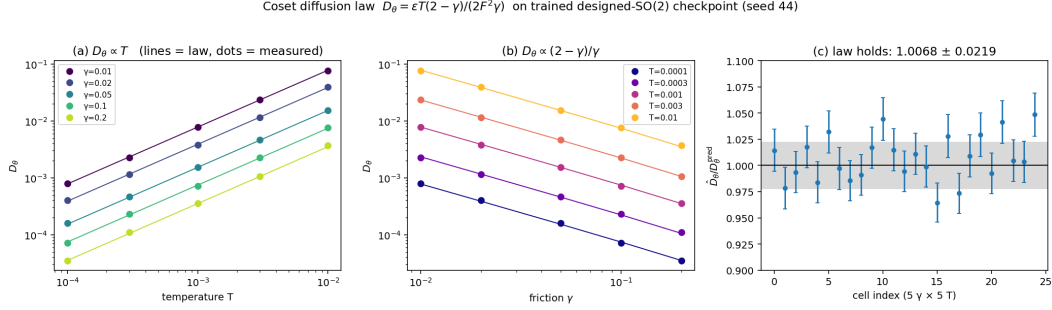


Figure 2: The coset diffusion law, panel by panel. (a) $D_\theta \propto T$ at five frictions; (b) $D_\theta \propto (2 - \gamma)/\gamma$ at five temperatures; (c) the ratio $\hat{D}_\theta / D_\theta^{\text{pred}}$ over all 25 cells, 1.0068 ± 0.0219 with the band drawn. *Multi-seed status: one trained designed-SO(2) checkpoint (seed 44), the single checkpoint on which the 25-cell grid was run; the five-seed statements of this appendix are the sign-flip and latch entries, not this grid. langevin_noise="fdt", Newtonian kinetic mode, $\Delta = 0.5$ rad. Designed testbed: verification.*

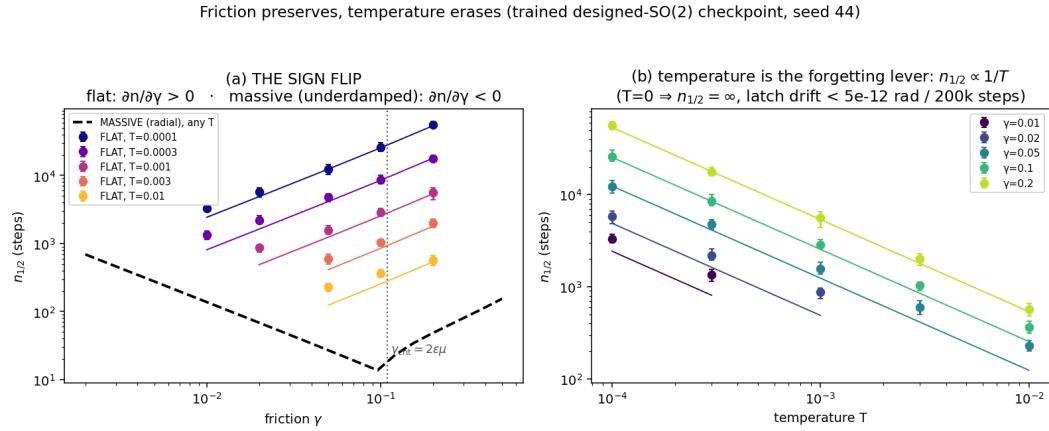


Figure 3: The sign flip, and the lever that does erase. (a) The flat coset's half-life *rises* with friction at every temperature, while the underdamped massive mode's falls — the two stored directions respond to the dial with opposite sign, with $\gamma_{\text{crit}} = 2\varepsilon\mu$ marked. (b) Temperature is the forgetting lever, $n_{1/2} \propto 1/T$ at every friction. *Multi-seed status: one trained designed-SO(2) checkpoint (seed 44); the 10/10 seed $\times$ temperature statement and the $+0.9552 \pm 0.0422$ slope are this appendix's five-seed numbers, not this figure's. langevin_noise="fdt", Newtonian kinetic mode, $\Delta = 0.5$ rad. Designed testbed: verification.*

The two regimes have different damping physics (exact one-step Jacobian at the vacuum)

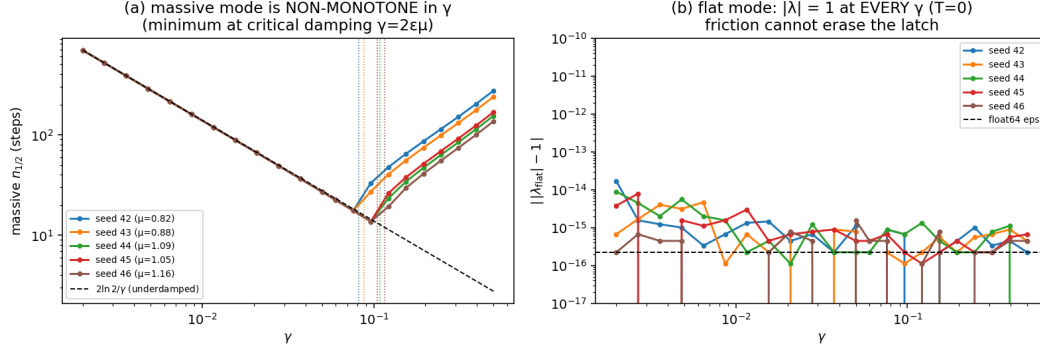


Figure 4: The two regimes, read off the exact one-step Jacobian at the vacuum. (a) The massive radial mode is non-monotone in γ , each seed's minimum sitting at its own $\gamma_{\text{crit}} = 2\varepsilon\mu$ (dotted), against the dashed underdamped envelope $2\ln 2/\gamma$ drawn on the same axes. (b) The flat mode holds $|\lambda| = 1$ at every γ to the float64 floor: friction cannot erase the latch. Five trained designed seeds, $T = 0$, $\dim 4$, hidden 64, $\varepsilon = 0.05$, laptop-CPU: verification.

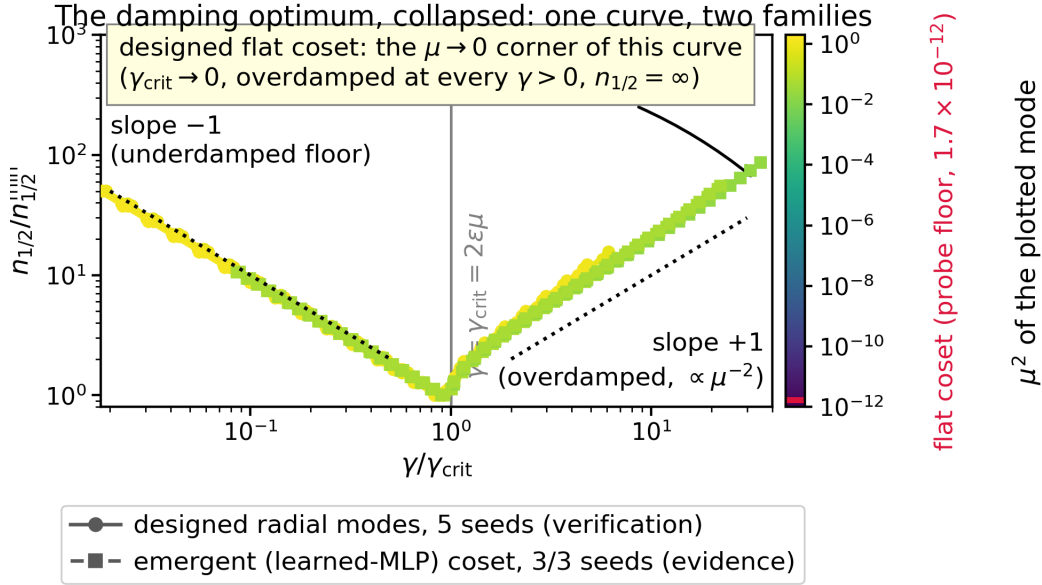


Figure 5: The damping optimum, collapsed — the full-size version of Figure 1. The same eight measured curves and the same ∓ 1 asymptotes, carrying the annotations the main-text version omits: the two-family legend, the $\gamma = \gamma_{\text{crit}} = 2\varepsilon\mu$ marker, the $\mu \rightarrow 0$ flat-coset callout, and the crimson probe-floor tick on the μ^2 colourbar (1.7×10^{-12}). No plotted value differs from Figure 1. (One-step Jacobian; $\dim 4$, hidden 64, $\varepsilon = 0.05$, laptop-CPU; `langevin_noise="fat"`, Newtonian kinetic mode.)

316 6.5×10^{-10}), so no seed's write cleared its washboard barrier. The pre-registered early-window
 317 clause fails, $\Gamma_{\text{jac}}/\Gamma_{\text{roll}}^{\text{early}} = 0.52\text{--}0.85$ against a band of $[0.65, 1.35]$; the amplitude sweep identifies
 318 that as mode mixing rather than anharmonicity, seed 44's early ratio being $0.539/0.527/0.522$ across
 319 the same $10\times$ range.

320 What the left branch does not contain. For $\gamma < \gamma_{\text{crit}}$ the coset eigenpair is complex with $|\lambda|^2 = 1 - \gamma$,
 321 so $n_{1/2} = \ln 2 / (-\frac{1}{2} \ln(1 - \gamma))$ independently of μ^2 and of the checkpoint: at $\gamma = 0.002$, emergent
 322 seeds 42 ($\mu^2 = 5.449 \times 10^{-2}$) and 43 ($\mu^2 = 2.029 \times 10^{-2}$) both print $\lambda_{\text{ret}} = 0.998999499$ and

Table 2: The shape claims on all four instruments ($\delta = 0.05$ rad, 48 γ values, 3 emergent seeds). The rollout rate instrument reproduces the Jacobian; both threshold instruments fail below γ_{crit} , so no rollout $n_{1/2}$ is ever quoted from a threshold there.

	γ_{crit}	argmin/ γ_{crit} (I-J / I-R1 / I-R2 / I-R3)	
seed 42	0.02334	0.8994 / 0.0857 / 0.2068 / 0.8928	-1.0023 / +0.0623
seed 43	0.01424	0.9046 / 0.1404 / 0.1404 / 0.9047	-1.0016 / +0.0623
seed 44	0.02265	0.9055 / 0.0883 / 0.3031 / 0.9028	-1.0022 / +0.0623
mean $\pm$ sd	$0.9032 \pm 0.0027 / 0.1048 \pm 0.0252 / 0.2168 \pm 0.0668 / 0.9001 \pm 0.0052$		

Table 3: The instrument gap is a level, not a rate: on the overdamped branch ($\gamma \geq 2\gamma_{\text{crit}}$) the decay *rates* agree while the *threshold* instrument is offset by a per-checkpoint constant flat in γ . Seed 44's full-window value (0.9216 ± 0.0027) sits 0.0004 below its pre-registered band because that window still contains a second, faster mode (early-window ratio 0.539); the window-split diagnostic converges monotonically to 1 into the asymptote. Cause of the threshold offset, measured: the ring write deposits only a fraction of its amplitude in the slow branch (envelope-fit intercept over $\delta = 0.638/0.644/0.327$).

seed	$\Gamma_{\text{jac}}/\Gamma_{\text{R3}}$ (asymptotic window)	$n_{\text{jac}}/n_{\text{R1}}$ (threshold)	CV of $n_{\text{jac}}/n_{\text{R1}}$ over the grid
42	0.9935	1.3307	2.2–3.7% over 21–25 γ -points
43	1.0000	1.7981	
44	0.9948	1.8204	
mean	0.995 ± 0.003	per seed, at $\delta = 0.05$ rad	$ d \log(\text{ratio})/d \log \gamma \leq 0.0034$ (threshold 0.10)

323 $n_{1/2} = 692.5$, against $2 \ln 2 / 0.002 = 693.1$. All model-dependent content of the V-curve is the
324 argmin location $\gamma_{\text{crit}} = 2\varepsilon\mu$ and the +1 branch; what the rollout adds is that the *nonlinear* model
325 obeys that identity, which was not guaranteed.

326 *Designed control, and the model-side bound.* Over 5 designed seeds $\times$ 48 $\gamma \times$ 3 write amplitudes $\times$
327 32 000 steps of direct $T = 0$ rollout, $\max_n |\theta(n) - \delta| = 9.400 \times 10^{-13}$ rad, so $\Gamma \leq 7.309 \times 10^{-17}$
328 per step and $n_{1/2} \geq 9.483 \times 10^{15}$ steps against a pre-registered bound of 10^{-6} rad; the matched
329 Hessian μ_{soft}^2 there is 1.60×10^{-15} , 1.52×10^{-15} , -2.92×10^{-16} , 2.94×10^{-16} , 1.21×10^{-16} .
330 *Collapse:* in $x = \gamma/\gamma_{\text{crit}}$, $y = n_{1/2}/\gamma_{\text{crit}}$ the three emergent curves agree to a median 1.15% on
331 I-J and 5.67% on I-R3; $y_{\text{min}} = 1.579/1.527/1.499$ and $1.631/1.535/1.662$ sit 22–24% below the
332 continuum $2\sqrt{2} \ln 2 = 1.961$ and $x_{\text{min}} \approx 0.90$ sits 27% above the continuum 0.707, both known
333 discrete-map corrections (matched designed control $x_{\text{min}} = 0.83$ –0.93). *Scope:* μ_{soft}^2 spans only
334 2.029×10^{-2} – 5.449×10^{-2} across the three seeds, a factor 2.7, so the eleven-decade span is one
335 decade of emergent variation attached to the designed $\mu \rightarrow 0$ corner and the rollout does not widen it.

336 *The emergent arm's own negative.* $1 - |\lambda_{\text{coset}}| = 1.06 \times 10^{-3}$ – 2.96×10^{-3} at $\gamma \approx 0.048$ against
337 a designed $\leq 1.1 \times 10^{-15}$, with a whole-grid emergent minimum of 7.6×10^{-5} – 2.0×10^{-4}
338 against a designed grid minimum of exactly 0. Written $\delta \in \{0.1, 0.3, 0.5\}$ rad relaxes completely
339 ($|\text{retained}| \leq 2.1 \times 10^{-3}$) where the designed checkpoint retains δ exactly; capacity is ≈ 1 –1.6
340 bits, not a continuum. Bias-corrected $n_{1/2}^{\text{obs}}/n_{1/2}^{\text{Goldstone}}$ at $\gamma = 0.2$, $\ell_\theta/\Delta < 0.06$, runs 2.75, 3.40
341 at $T = 2 \times 10^{-3}$ to 0.94–1.25 at $T \geq 4 \times 10^{-3}$, giving $T^* \approx 3 \times 10^{-3}$ (predicted 2.72–3.66 $\times$
342 10^{-3}) against a washboard barrier of 2.29 – 3.57×10^{-2} . *Reporting rule:* raw exponents do not
343 discriminate — a matched designed control gives $d \log n_{1/2}/d \log T = -0.53, -0.60, -1.04$ and
344 $d \log n_{1/2}/d \log \gamma = +0.78, +0.63, +0.55$, indistinguishable from emergent, because both families
345 carry the ℓ_θ/Δ boundary layer (up to 2.03). Only $|\lambda_{\text{coset}}|$ and bias-corrected deep-diffusive cells
346 discriminate.

347 C The friction-hole vault, on both arms

348 *Designed SO(2) plus an analytic field, 3 seeds $\Rightarrow$ verification; emergent MLP, 3 seeds $\Rightarrow$ evidence.*
349 *Pre-registered on both arms. All cells `langevin_noise="fdt"`, Newtonian kinetic mode, $\Delta = 0.5$
350 rad.*

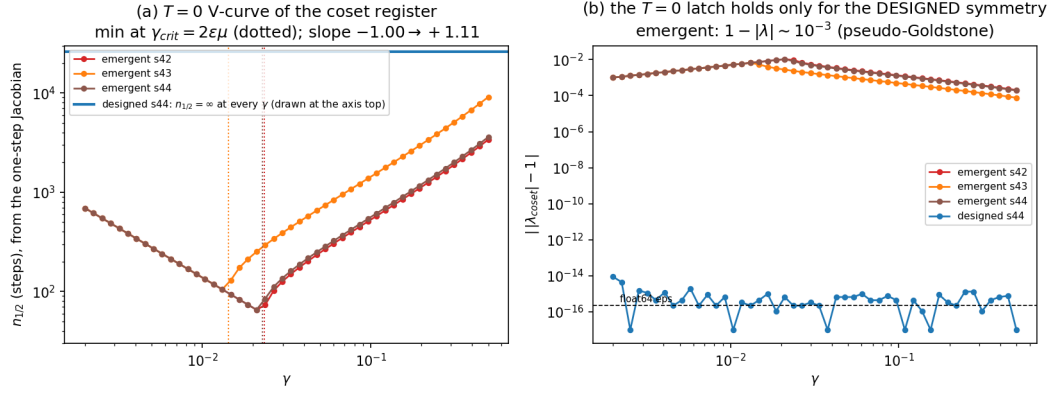


Figure 6: The emergent V-curve un-collapsed, and the negative beside it. (a) The $T = 0$ half-life against friction per emergent seed from the one-step Jacobian, each with its own $\gamma_{crit} = 2\epsilon\mu_{soft}$ dotted; the designed coset has no finite half-life at any γ and is drawn at the axis top. (b) $||\lambda_{coset}|| - 1||$: exact to the float64 floor for the designed symmetry and short by $\sim 10^{-3}$ on every emergent seed, which is the pseudo-Goldstone statement of this section in one panel. *Three emergent MLP seeds plus a matched designed control: evidence.*

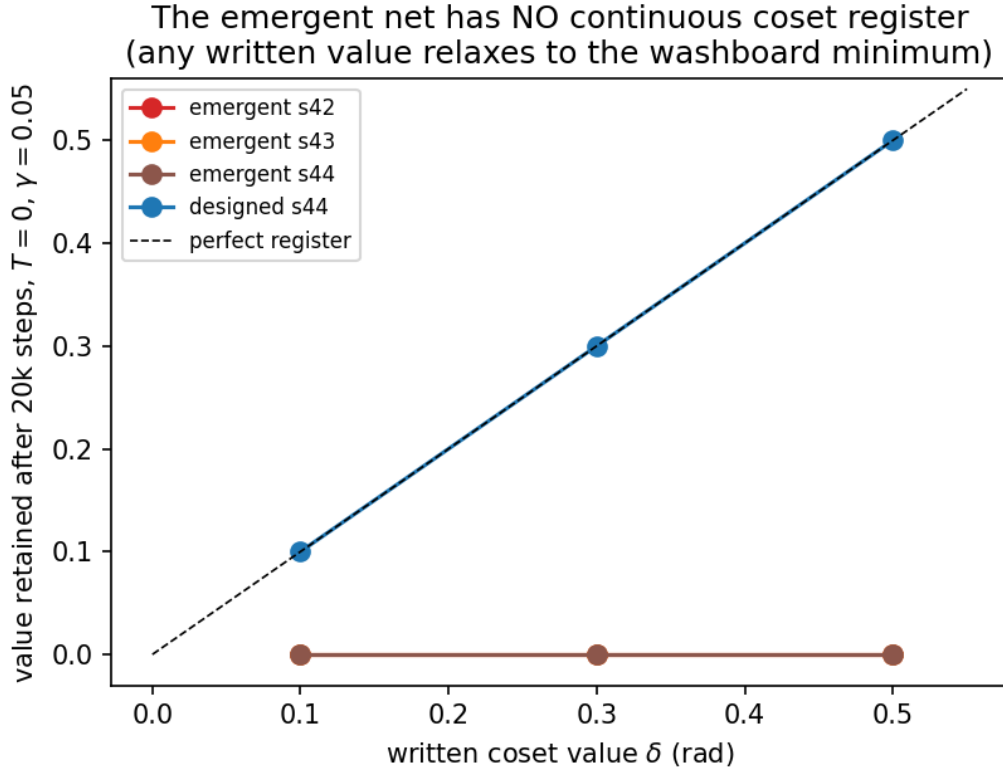


Figure 7: The emergent unit has no continuous coset register. A written coset value δ is held for 20 000 steps at $T = 0$, $\gamma = 0.05$: on every emergent seed the value relaxes to the washboard minimum and nothing is retained, where the designed checkpoint returns δ exactly. *Three emergent MLP seeds plus a matched designed control: evidence.* This is the figure behind the capacity statement of ≈ 1 –1.6 bits.

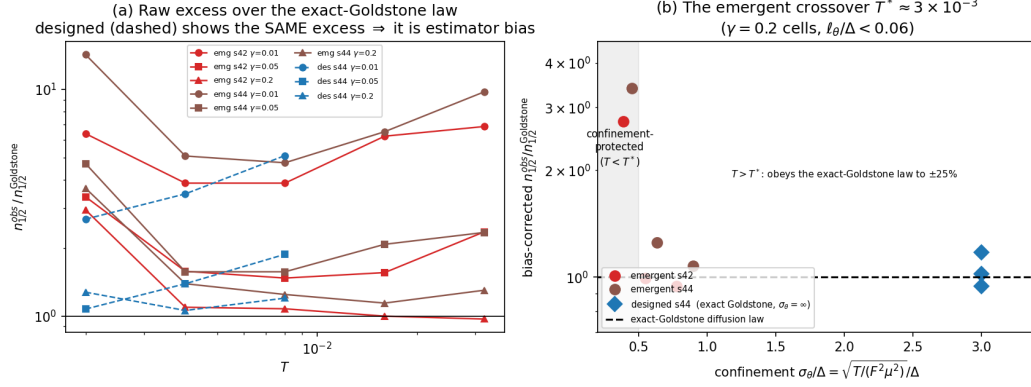


Figure 8: The crossover T^* , and why the raw curve cannot be read directly. (a) The raw excess over the exact-Goldstone law is the same on the matched designed control as on the emergent seeds, so the excess is estimator bias rather than physics. (b) Bias-corrected, the emergent coset joins the exact-Goldstone law above $T^* \approx 3 \times 10^{-3}$, the shaded band being the confinement-protected cells below it. *Multi-seed status, stated plainly: two emergent seeds and one matched designed control, an $n < 3$ cell.* $\Delta = 0.5$ rad, $\ell_{\theta}/\Delta < 0.06$, langevin_noise="fdt", Newtonian kinetic mode: evidence.

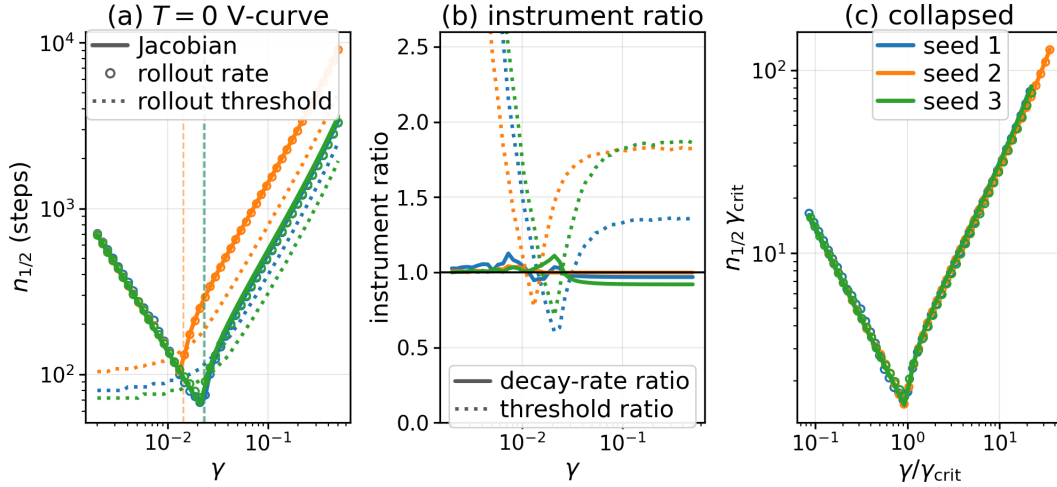


Figure 9: The V-curve measured a second way. (a) $T = 0$ $n_{1/2}(\gamma)$ per emergent seed on the one-step Jacobian (I-J, lines), the rollout envelope-rate instrument (I-R3, circles) and the rollout first-crossing threshold (I-R1, dotted), at $\delta = 0.05$ rad. (b) The per- γ instrument ratio: the rate ratio sits on 1, the threshold ratio is offset and flat in γ above γ_{crit} . (c) The same data in collapsed variables. (*Emergent MLP, 3 seeds, dim 4, hidden 64, $\varepsilon = 0.05$, float64, $T = 0$, pre-registered $\Rightarrow$ evidence.*)

351 *Mechanism.* The shipped field is absorb-only, $p \leftarrow (1 - \gamma_{\phi})(1 - \gamma)p + \sigma(\gamma)\xi$ with $\sigma^2 = MT\gamma(2 - \gamma)$
352 set by the scalar γ , so with $\gamma_{eff} = 1 - (1 - \gamma)(1 - \gamma_{\phi})$, $T_{local} = T\gamma(2 - \gamma)/[\gamma_{eff}(2 - \gamma_{eff})]$ and
353 $D_{\theta} = \varepsilon T\gamma(2 - \gamma)/(2F^2\gamma_{eff}^2)$: a vault of $(\gamma_{eff}/\gamma)^2$. A coupled bath would give $D_{\theta} \propto \gamma_{eff}^{-1}$ and a
354 vault of $[\gamma_{eff}(2 - \gamma)]/[\gamma(2 - \gamma_{eff})] = 13.88 \times$ at $\gamma = 0.05$, $\gamma_{\phi} = 0.5$. The hypotheses differ by
355 $7.942 \times$; the coupled-bath value is a refuted prediction and never the vault.

356 *Designed arm.* $\text{Var}(p_i)/(M_i T) = 0.9986$ ($\gamma_{\phi} = 0$), 0.36171 ± 0.00129 (0.1), 0.23044 (0.2),
357 0.17513 (0.3), 0.12600 ± 0.00031 (0.5; absorb prediction 0.12591), so $T_{local} = 1.26 \times 10^{-4}$, a
358 $7.94 \times$ refrigerator. $\text{MSD}/\text{pred}_{\text{absorb}} = 1.0011 \pm 0.0215$ over 15 cells; the D_{θ} -based vault is
359 $8.42 \times$, $22.39 \times$, $44.31 \times$, $107.77 \pm 4.78 \times$ at $\gamma_{\phi} = 0.1/0.2/0.3/0.5$ (absorb prediction 110.25). The
360 direct first-passage vault reads $86.97 \pm 2.94 \times$ and is boundary-layer biased on the outside arm
361 ($\ell_{\theta}/\Delta = 0.079$), so $107.77 \times$ is the quoted number and travels with its estimator's name. Scalar

control at the same γ_{eff} : $13.28 \pm 0.12 \times$ with $\text{Var}(p) = M_i T$ intact, field/scalar 8.11 ± 0.37 against a predicted 7.942. At $T = 0$ the hole erases nothing (drift $\leq 1.75 \times 10^{-12}$ rad per 200k steps inside and outside, 12/12 cells; $||\lambda|_{\text{max}} - 1| \leq 2.0 \times 10^{-15}$ over 27 cells including the hole edge) and attenuates the *write* instead: $\theta_{\text{settle}}(\text{in})/\theta_{\text{settle}}(\text{out}) = 0.09576$ against the transport law's 0.095238.

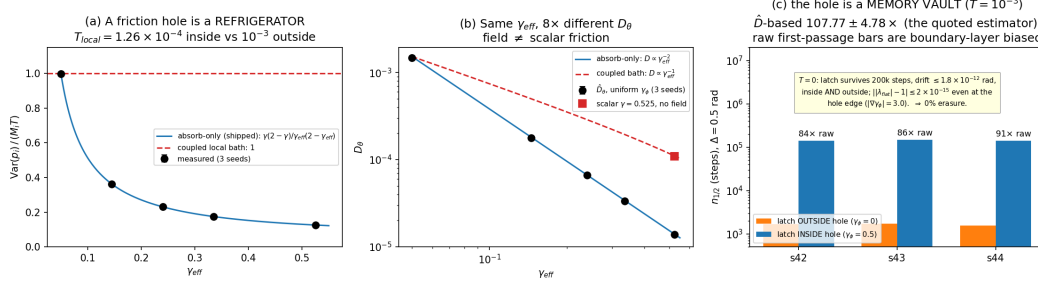


Figure 10: The friction hole as a memory vault, on the designed arm. (a) The hole is a refrigerator: the momentum variance follows the absorb-only curve and rejects the coupled-bath prediction, giving $T_{\text{local}} = 1.26 \times 10^{-4}$ inside against 10^{-3} outside. (b) At the same γ_{eff} the diffusion constant differs by $8 \times$ between a friction field and a scalar friction, which is the mechanism statement in one panel. (c) The vault factor: $107.77 \pm 4.78 \times$ from the $\hat{D}_\theta$ estimator, the quoted number; the per-seed bars beneath it are raw first-passage ratios, boundary-layer biased on the outside arm, which is why the pooled first-passage reading of $86.97 \pm 2.94 \times$ is not quoted as the vault. The inset states the $T = 0$ control: the hole erases nothing. *Designed $SO(2)$ with an analytic field, 3 seeds, $T = 10^{-3}$, $\Delta = 0.5$ rad, $\text{langevin_noise} = \text{"fdt"}$, Newtonian kinetic mode, laptop-CPU: verification.*

Emergent arm. The coset there is a bounded mode ($\mu_{\text{adiab}}^2 = 2.77 \times 10^{-2} - 8.93 \times 10^{-2}$), so the designed block estimator is biased low by up to 55% and is replaced by a window-free Ornstein-Uhlenbeck fit $\text{MSD}(n) = A(1 - e^{-n/\tau})$, $\hat{D} = A/(2\varepsilon\tau)$, cross-validated on designed before use ($112.58 \pm 1.09 \times$ against the banked $107.77 \pm 4.78 \times$; $14.16 \pm 1.38 \times$ against $13.28 \pm 0.12 \times$; 8.03 ± 0.80 against 8.11 ± 0.37).

Table 4: The refrigerator law transfers exactly to an emergent register: $\text{Var}(p_i)/(M_i T)$ against the absorb-only prediction, 2048 walkers $\times$ 40 samples, 3 emergent seeds; over all 24 field cells observed/absorb = 0.9998 ± 0.0019 , and at $\gamma_\phi = 0.5$ $T_{\text{local}}/T = 0.12570$, a $7.955 \times$ refrigerator against 7.942 predicted. The coupled bath predicts 1.0 everywhere and is measured at 0.2235.

T	γ_ϕ	γ_{eff}	absorb pred.	coupled pred.	measured	obs/absorb
4×10^{-3}	0.0	0.050	1.00000	1.0	0.99815 ± 0.00257	0.9981
4×10^{-3}	0.1	0.145	0.36249	1.0	0.36326 ± 0.00067	1.0021
4×10^{-3}	0.2	0.240	0.23082	1.0	0.23062 ± 0.00080	0.9991
4×10^{-3}	0.3	0.335	0.17480	1.0	0.17473 ± 0.00046	0.9996
4×10^{-3}	0.5	0.525	0.12591	1.0	0.12592 ± 0.00055	1.0001
8×10^{-3}	0.5	0.525	0.12591	1.0	0.12548 ± 0.00032	0.9966

The diffusion law on the emergent arm. On bounded cells, $\hat{D}_{\text{ou}}/D_{\text{absorb}} = 1.0480$ ($\gamma_\phi = 0, 1$ cell), 1.1026 ± 0.0799 (0.1, 4), 1.0158 ± 0.0519 (0.2, 5), 1.0552 ± 0.0496 (0.3, 5), 1.0418 ± 0.0519 (0.5, 6); against the coupled bath the same cells give 0.1312 ± 0.0065 at $\gamma_\phi = 0.5$. Law-referenced vault $106.1 \pm 5.0 \times$ (6 cells; prediction $110.25 \times$; designed $107.77 \pm 4.78 \times$). A measured/measured ratio over the same cells reads $297.8 \pm 196.8 \times$ (median $234.4 \times$); it is recorded and is never the vault number, because 5 of the 6 $\gamma_\phi = 0$ cells fail the bounded test ($\sigma_\theta^{\text{ou}} = 0.42 - 885$ rad against a Boltzmann $0.24 - 0.65$), while every $\gamma_\phi \geq 0.2$ cell passes.

The contrast number is designed-only, and the failure mode is the control. A scalar γ does not cool — $\text{Var}(p)/(MT) = 0.99288 \pm 0.02354$ at $\gamma = 0.525$ with no field, where both hypotheses predict 1.0, against 0.126 for the field at the same γ_{eff} — so the mechanism statement (a friction field is not locally stronger friction; it is locally stronger friction *plus* a local refrigerator) stands on emergent. The contrast measures 23.39 ± 10.06 (per-cell $9.14/25.75/26.96/21.70/41.31/15.48$)

Table 5: The hole confines an emergent register, and the laundering control shows friction alone does not. $T = 4 \times 10^{-3}$, 1024 walkers, burn-in 20 000–80 000 steps, spread by interquartile range; “hop” means $|\theta| > 1$ rad. The pre-registered stationary-spread comparison is *void* here and no in/out spread ratio may be quoted from these checkpoints: the control that must return exactly 1.000 returns 0.4586 ± 0.1181 .

seed	arm	γ_{eff}	σ_θ (IQR)	hop fraction	Boltzmann at T_{local}	obs/pred
42	no hole, $\gamma = 0.05$	0.050	0.40795	5.50%	0.24073	1.6946
42	scalar 0.525 (control)	0.525	0.20034	0.73%	0.24073	0.8322
42	γ_ϕ hole	0.525	0.06442	0.0000	0.08542	0.7542
43	no hole	0.050	1.17601	42.97%	0.46059	2.5533
43	scalar 0.525	0.525	0.35335	10.20%	0.46059	0.7672
43	γ_ϕ hole	0.525	0.08628	0.0000	0.16343	0.5279
44	no hole	0.050	0.36922	2.36%	0.24034	1.5363
44	scalar 0.525	0.525	0.21574	0.26%	0.24034	0.8977
44	γ_ϕ hole	0.525	0.07279	0.0000	0.08528	0.8535

383 against a pre-registered band $[6.5, 9.5]$ and a falsifier at > 14 : it fired. The cause is the control arm,
384 whose own $\hat{D}_{\text{ou}}/D_{\text{flat}}$ is $1.1768/3.7319/3.3782/2.7999/5.2284/2.0506$ — at $\gamma = 0.525$ with no
385 cooling the emergent coset is still anharmonically delocalised — while the field arm sits on the absorb
386 law at 1.02 – 1.15 . The *direction* transfers; the *number* 8.11 ± 0.37 is a designed quantity.

387 *First passage, and the confound it exposed.* An emergent potential has no rotational symmetry, so a
388 $\theta = \pi$ “outside” arm is not a vacuum: $|\nabla V| = 0.0420/0.1049/0.0414$ there against 0.0000 at $\theta = 0$,
389 $V(\pi) - V(0) = +0.03163/ +0.06003/ +0.04315$, and μ_{adiab}^2 differing by up to $2.15\times$ between
390 the sites. Any emergent first-passage number quoted against a $\theta = \pi$ baseline is confounded and
391 a same-site baseline is required. Same-site vaults ($|\Delta\theta| \geq 0.5$ rad, $T = 4 \times 10^{-3}$, $\gamma = 0.05$, 256
392 walkers, hole radius 1.0): $> 1379\times$ (seed 42), $35.5\times$ (43), $> 1290\times$ (44), the first and third being
393 censoring-limited *lower bounds* (86.7% and 93.4% censored at a 10^6 -step cap; closing them needs
394 $\approx 10^7$ steps per walker). The pre-registered $> 300\times$ holds 2/3 and the falsifier fires on seed 43,
395 already flagged as geometrically atypical — softest μ^2 , shallowest barrier (6.8×10^{-3}), coset at
396 $\sigma_\theta/\Delta = 0.92$ outside and 0.33 inside, never leaving the diffusive regime. The confound’s signature
397 is seed 44, whose $\theta = \pi$ first passage (15585) is $20\times$ its own $\theta = 0$ value (775) and whose scalar
398 control reads $0.27\times$: more friction apparently *speeding* escape, impossible for a stationary register.

399 *Scope carried by the whole emergent arm.* Three model seeds; a single PRNG stream per cell, so
400 every spread is across model seeds, not noise realizations; both temperatures above T^* while the
401 designed study’s trustworthy-cell restriction is $T \leq 3 \times 10^{-3}$, so every measurement is squeezed
402 between T^* and delocalisation and only cells passing the bounded test support a law claim. The
403 hole is a ball of radius 1.0 in the full 4-D space, so excursions of order 1.0 ($\approx 4.5\sigma_r$) leave it — a
404 plausible contributor to seed 43’s low vault, not separately measured. No temperature field was built.

405 D Exact store-level deletion: prior art and the measured tables

406 *Designed atom store, dim 3, capacity 8–64, no learning anywhere $\Rightarrow$ verification of the mechanism’s*
407 *exactness. Every deletion statement carries its scope: store level only, under the three stated*
408 *conditions, with recency-based eviction permanently excluded.*

409 *Prior art, in full.* Uniquely represented (canonical) data structures date to Snyder (1977) and
410 Andersson & Ottmann (1995); *obliviousness*, a distributional condition on the memory representation
411 which explicitly does not require canonical representation, is Micciancio’s (1997); the weak and
412 strong history-independence notions used here are Naor & Teague’s (2001); the open-addressed
413 realisation is Blleloch & Golovin’s (2007), whose table is a stable matching between keys and
414 slots under a global key priority, and it is that priority-greedy rule and that delete-time fix-up
415 cascade we use; Blleloch, Golovin & Vassilevska (2008) had already taken unique representation
416 into computational geometry. We claim no priority over order-independent placement and no novelty
417 for the displacement rule or its delete-time repair. New is the composition: slots that are a lattice
418 packing, so the canonical representation carries a minimum-separation certificate; content that is a
419 continuously decaying amplitude whose decay commutes with deletion; a canonical object that is

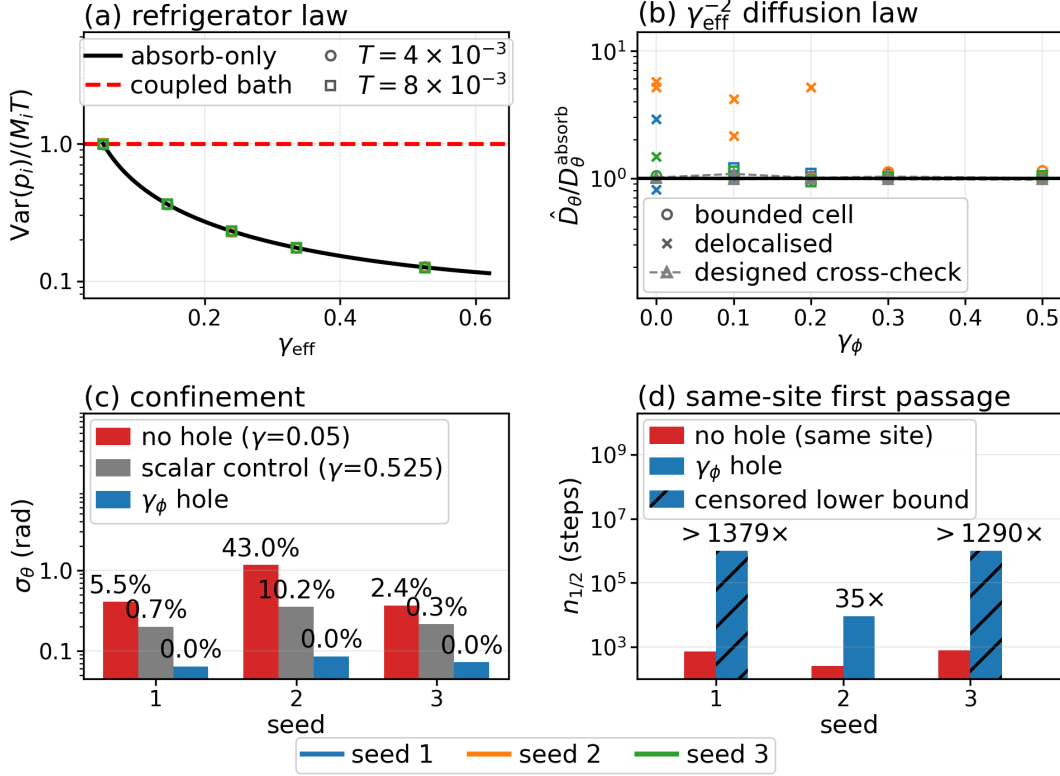


Figure 11: The vault on an emergent register; panels are labelled by the pre-registered prediction each tests. (a) The refrigerator law, $\text{Var}(p_i)/(M_i T)$ against γ_{eff} for 3 seeds $\times$ 2 temperatures, on the absorb-only curve with the coupled-bath prediction (dashed) rejected. (b) The γ_{eff}^{-2} diffusion law: bounded cells as circles and squares, delocalised cells as crosses, the designed cross-check overlaid. (c) Confinement: stationary spread and hop fraction for no hole, the scalar control and the hole. (d) Same-site first passage; hatched bars are censored lower bounds. (*Emergent MLP, 3 seeds, $T \in \{4, 8\} \times 10^{-3}$, both above T^* , $\Delta = 0.5$ rad $\Rightarrow$ evidence.*)

an energy function a dynamical read relaxes into; and a price that is negative where strong history independence is known to cost as much as an exponential slowdown relative to the weak notion for comparison-based structures (Buchbinder & Petrank, 2003). “Fix-up cascade” is our name for the NEXT-chain delete-time repair of their Figure 1; the algorithm is theirs either way.

Exactness, measured at zero tolerance. 24 exhaustive orders at $n = 4$, 200 random orders at $n \in \{8, 16, 40, 64\}$, 100 write/delete interleavings and mid-decay deletes: all bit-identical, minimum spacing exactly 1.540000; on the shipped store the centres, payloads, amplitudes and active mask are byte-equal and $V(q)$ is byte-equal on 64 random queries. *The packing price is negative:* at $K = 64$ the canonical rule admits 61/64 offered keys deterministically ($\sigma = 0$), per-offered 0.9531, rising to 64/64 when the sizing is inflated by $\times 1.05$, where the stochastic refuse-and-relocate allocator admits 43/64 on the same seeds (per-offered 0.6719); delete-time churn is 2.836 moves per delete at full load. *At overflow, without the waitlist, exactness fails:* with 7 cells and 8 offers, $\text{AUC}(n_{\text{live}}) = 1.00000 \pm 0.0000$, white-box address-depth $\text{AUC} = 0.91409 \pm 0.0813$, byte-equal fraction 0.0000. A pre-registration that failed informatively: $\text{AUC}(z_{\text{hole}}) \geq 0.85$ was registered there and measured 0.576, because the fix-up cascade *fills* the freed cell with a re-placed survivor.

What decays and what does not. Against an exact adversary the white-box per-example AUC is 1.000 at all 18 amplitude levels down to the floor; what decays is the effect size, the depth gap being $0.3935 \cdot A$ to four decimals at every level (the pre-registered closed form $1 - e^{-1/2}$) and d' falling $1560 \rightarrow 79.6$, exactly linear in A — the pre-registered falsifier fired on the AUC metric, as registered. Against a resolution-limited adversary the grading is real and the pre-registered law survives, $A_{75} = 2.57\sigma_{\text{obs}}$ measuring $0.0780/0.2627/0.7537$ at $\sigma_{\text{obs}} = 0.03/0.1/0.3$ against predicted $0.0771/0.2570/0.7710$;

Table 6: The laundering control that fires. Designed store, shipped two-phase read, no learning anywhere; 8 targets $\times$ 3 seeds, 128 paired worlds per target, 18 amplitude levels. Against an *exact* adversary a boolean TTL flag inside the same store is as detectable as an $e^{-\lambda t}$ amplitude, so there is no measurable differentiator at the AUC level; the control fails to fire only against a resolution-limited adversary, and that resolution is our own modelling choice — which is why the main text leads with the retrieval geometry instead.

adversary	physical decay	TTL flag in the same store	separation
white-box, exact	1.000	1.000	0.000
query-level membership, exact	0.983	1.000	0.017
query-level membership, $\sigma_{\text{obs}} = 0.1$	0.559	0.996	0.437

441 but σ_{obs} is our modelling choice, so its defensible use is mechanistic, never protective. *State the*
442 *|c| distribution with any retention number:* every retention magnitude here is for the controller-
443 placed disk (proposal radius 2.2869, packing bound 8.00); on a max-radius ring the same store gives
444 retention 0.500 at $A = 0.06$ where this placement gives 0.886. *Retrieval geometry:* R_{50} contracts
445 1.146/1.083/0.979/0.874/0.752 as $A : 1 \rightarrow 0.5 \rightarrow 0.2 \rightarrow 0.1 \rightarrow 0.06$, matching the pre-registered
446 saddle prediction at all five points (± 0.20), where a TTL vector store’s lookup radius is a constant
447 hard step at 0.75–0.77.

448 *The trilemma, and what it costs.* The shipped store drops amplitude-independent address hold, which
449 is why effective lifetime correlates $r = -0.85$ with a_i^2 ; the recommended gated-stiffness channel
450 drops amplitude-independent latency and is the compute-adaptive-read dial — and it is *not* the
451 shipped arm. Both previously proposed repairs are refuted: multiplying payloads by amplitudes
452 kills the value criterion at $A = 1 - \text{tol}/|a_i|$ (measured death amplitudes 0.90/0.80/0.70/0.30
453 against the formula’s 0.900/0.860/0.767/0.300) and inverts the payload dependence; launching
454 the payload coordinate at the address-mapped point breaks the anti-decoration guard, after which a
455 trivial substitute passes 0.672 of queries with no dynamics at all. Behind its flag the gated channel
456 contracts R_{50} 1.135 $\rightarrow$ 0.771, retention std 0.274 $\rightarrow$ 0.016, worst codeword 0.398 $\rightarrow$ 0.972; the
457 registered payload-independence test *fails as a correlation* ($|r|$ only 0.846 $\rightarrow$ 0.667) and passes as
458 an effect size; and at $\gamma_{\text{read}} = 0$ it is oscillatory, period $2\pi/\sqrt{\kappa(g_0 + A)}$ (log-log slope -0.5264),
459 so no settling law may be quoted without a nonzero read friction. *No deletion-cost statement is*
460 *made:* the shipped canonical synchronisation is an $O(n)$ rebuild per operation and the amortized-cost
461 experiment has not been run.

462 E Prominent negatives

463 *Every negative below is on the record with its measurement; none is dropped, and each anchors a*
464 *future-work item.*

claim tested	result
A continuous coset register emerges from training	No: an MLP potential on symmetric data yields a mid-spectrum massive mode ($1.7\text{--}4.9\times$ softer than the stiffest), capacity $\approx 1\text{--}1.6$ bits, complete relaxation of any written value. The $T = 0$ latch is designed-only.
Raw T/γ -exponents discriminate designed from emergent	No: a matched designed control reproduces the same shallow slopes; only ℓ_θ/Δ -controlled cells and $ \lambda_{\text{coset}} $ discriminate. Never an $n_{1/2}$ without Δ and ℓ_θ/Δ .
The friction hole thermalizes locally (coupled bath)	Rejected by a pre-registered measurement, by a factor 8.11: the field is absorb-only and the vault is $107.77 \pm 4.78\times$, against $13.28 \pm 0.12\times$ for the scalar control.
A friction field can <i>forget</i> a coset	No: at $T > 0$ it <i>lengthens</i> coset memory. Erasing latched content needs a temperature or tilt field — which explains the observed failure of a learned friction field to recover fit on a stationary vacuum.
Our own sign-flip prediction, in both observables	Falsified in one: a register written at the washboard minimum cannot flip back, and both signs coexist on one checkpoint for different observables. A sign-flip figure that does not name the initial condition is meaningless.
Amplitude decay reduces an exact adversary's distinguishability	No, at any point in an item's life: white-box AUC 1.000 at all 18 amplitude levels; what decays is the effect size. The graded curve exists only relative to a stated adversary resolution, our own modelling choice.
Canonical placement is exact at overflow without the waitlist	No: $\text{AUC}(n_{\text{live}}) = 1.000$, byte-equal fraction 0.0000. Every deletion sentence carries its scope clause; recency eviction is permanently excluded.
The placement algorithm is ours	No: Blleloch & Golovin (2007) own it outright, and unique representation in a geometric setting is also taken (2008). Our claim is the composition only.
A per-deletion cost claim can be made	No: the shipped synchronisation is an $O(n)$ rebuild per operation, so the amortized-cost formalism cannot be instantiated on our own code until that experiment runs.
The payload-lifetime coupling has a repair	Both proposed repairs are refuted, and the registered payload-independence statistic fails as a correlation while passing as an effect size. At zero read friction the gated channel is oscillatory, not relaxational.

claim tested	result
A pseudo-Goldstone tilt is a lifetime dial on a learned store	Refuted <i>in sign</i> : a tilt monotonically <i>reduces</i> the smallest eigenvalue on two independent implementations and every family, because a written site is not a common vacuum of its own atoms. The relation survives only in the single-atom geometry it was specified in.
A tilt sets the soft scale on a learned store	No: the confinement floors the soft eigenvalue at 2α , capping lifetime at $\tau_{\max} = \Gamma/2\alpha$. The registered damped-mode floor is confirmed, the $1/\text{tilt}$ branch unreachable, and lowering the confinement breaks the write.
The lifecycle's protected-fraction leg is exercised	No: 0 refusals on the stream at the measured operating point. Reported as <i>unexercised</i> — not working and not broken.
A task-level payoff is claimed	No: forgetting here is shown predictable, controllable, removable and schedulable, not shown to improve a downstream metric. External benchmarks won on their own headline metric = ZERO.
The field/scalar vault contrast transfers to an emergent register	No: 23.39 ± 10.06 against 8.03 ± 0.80 from the identical estimator on designed; the pre-registered falsifier fired. The cause is the laundering control itself, off its own law by $1.18\text{--}5.23\times$. The number is designed-only; its direction transfers.
468 The emergent stationary-spread diagnostic is measurable above T^*	No: its own control, which must return 1.000, returns 0.4586 ± 0.1181 — the no-hole arm is a hopping rotor, so no in/out spread ratio may be quoted. The confinement statement replaces it.
A $\theta = \pi$ baseline is valid for emergent first passage	No: $ \nabla V = 0.042\text{--}0.105$ there against 0.0000 at $\theta = 0$, and one seed's scalar control reads $0.27\times$, i.e. more friction apparently speeding escape. Same-site baselines are required; the designed arm is protected by symmetry but was never checked, and that check is owed.
Threshold instruments are usable below γ_{crit}	No: a first-crossing $n_{1/2}$ puts its argmin at the grid edge with a below-slope of $+0.0623 \pm 0.0120$ where the law says -1 , and last-crossing carries an additive half-period bias. Only the envelope-rate instrument is quotable there.
Our microscopic explanation of the instrument offset	Wrong, though the offset itself is real, constant and quotable: the pre-registered single-slow-mode projection fails (amplitude fraction $0.638/0.644/0.327$ does not order like the angular overlaps) and the early-window ratio $0.52\text{--}0.85$ is amplitude-independent — multi-mode transient, not anharmonicity.
The emergent first-passage vault is measured	Only bounded: $> 1379\times$ and $> 1290\times$ at $86.7\%/93.4\%$ censoring, and $35.5\times$ on the geometrically atypical seed. Three seeds cannot say which is typical.

469 *Anonymization note.* Any supplementary or linked material, including code, is anonymized; only an
470 anonymized snapshot may be linked.